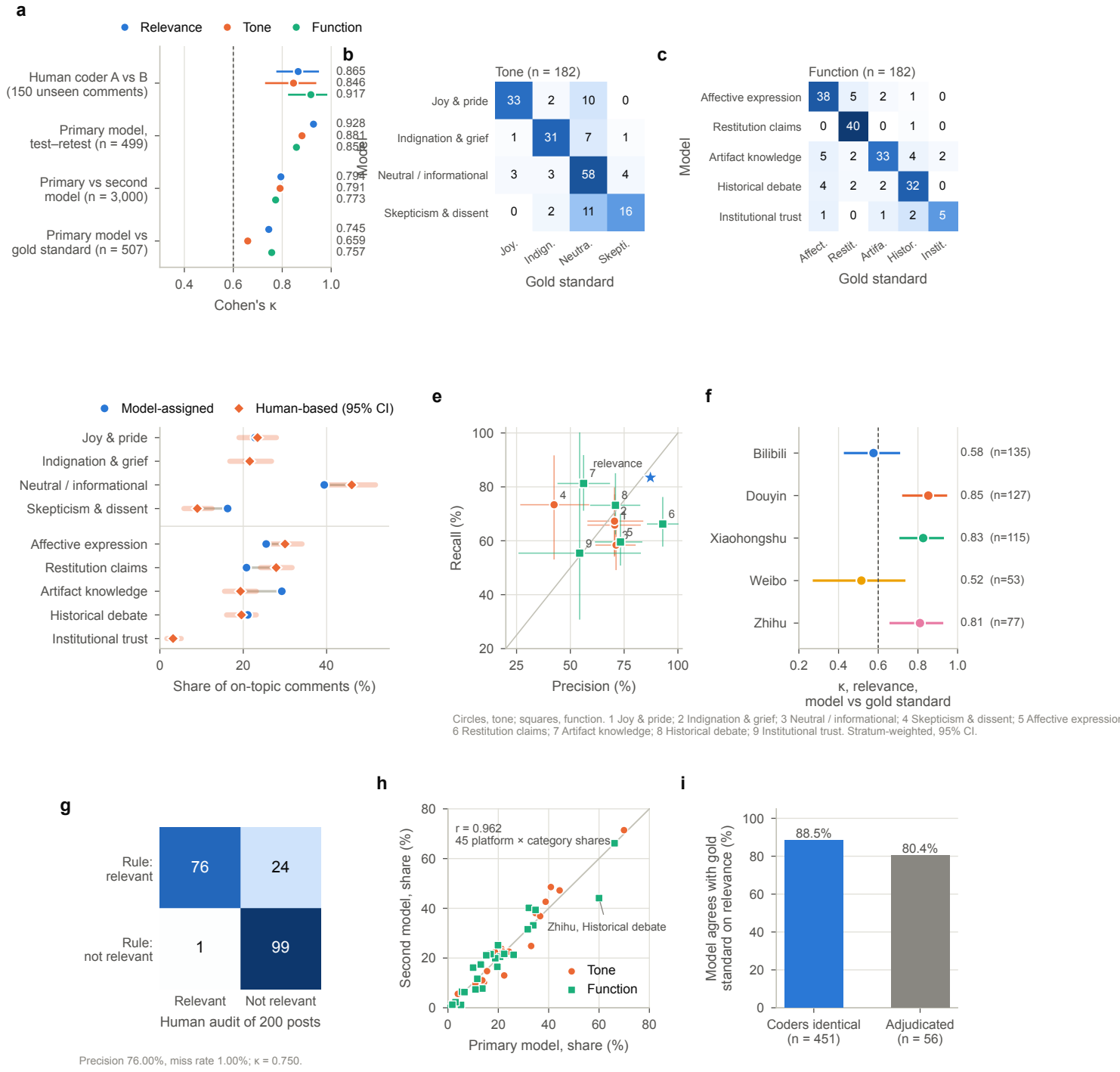

a
Relevance
Tone
Function
Human coder A vs B (150 unseen comments)
Primary model, test–retest (n = 499)
Primary vs second model (n = 3,000)
Primary model vs gold standard (n = 507)
0.865
0.846
0.917
0.928
0.881
0.858
0.794
0.791
0.773
0.745
0.659
0.757
0.4
0.6
0.8
1.0
Cohen's κ
b
Tone (n = 182)
Joy & pride
Indignation & grief
Neutral / informational
Skepticism & dissent
33 2 10 0
1 31 7 1
3 3 58 4
0 2 11 16
Joy.
Indign.
Neutra.
Skepti.
Model
Gold standard
c
Function (n = 182)
Affective expression
Restitution claims
Artifact knowledge
Historical debate
Institutional trust
38 5 2 1 0
0 40 0 1 0
5 2 33 4 2
4 2 2 32 0
1 0 1 2 5
Affect.
Restit.
Artifa.
Histor.
Instit.
Model
Gold standard
d
Model-assigned
Human-based (95% CI)
Joy & pride
Indignation & grief
Neutral / informational
Skepticism & dissent
Affective expression
Restitution claims
Artifact knowledge
Historical debate
Institutional trust
0
20
40
Share of on-topic comments (%)
e
relevance
Recall (%)
Precision (%)
20
40
60
80
100
25
50
75
100
f
Bilibili
Douyin
Xiaohongshu
Weibo
Zhihu
0.58 (n=135)
0.85 (n=127)
0.83 (n=115)
0.52 (n=53)
0.81 (n=77)
0.2
0.4
0.6
0.8
1.0
κ, relevance, model vs gold standard
Circles, tone; squares, function. 1 Joy & pride; 2 Indignation & grief; 3 Neutral / informational; 4 Skepticism & dissent; 5 Affective expression; 6 Restitution claims; 7 Artifact knowledge; 8 Historical debate; 9 Institutional trust. Stratum-weighted, 95% CI.
g
Rule: relevant
Rule: not relevant
76
24
1
99
Relevant
Not relevant
Human audit of 200 posts
Precision 76.00%, miss rate 1.00%; κ = 0.750.
h
r = 0.962
45 platform × category shares
Zhihu, Historical debate
Tone
Function
Second model, share (%)
Primary model, share (%)
0
20
40
60
80
i
88.5%
80.4%
Model agrees with gold standard on relevance (%)
0
20
40
60
80
100
Coders identical (n = 451)
Adjudicated (n = 56)